Monitoring Economics to enhance Economic Tasks to forecast Crises using Digital Technologies with Multimodal Deep Learning, Education and Safe Information, in Sustainability

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Abstract

This study examines smart education in monitoring the economy and preventing crises, knowing that education plays a leading role in awareness raising, crisis absorption, and general human security, in sustainability. It presents a multi-modal deep learning model based on CNN-LSTM while discussing integrating artificial intelligence (AI), Internet of Things (IoT), smart data, social networking, and digital technologies in economic monitoring and crisis prevention through smart education to increase digitization levels and well succeed in digital transformation. This multimodal model improves economic activity by providing interactive and intelligent education to keep people, especially managers, out of an ongoing event to make quick decisions. The study has a solid theoretical basis and demonstrates rigor and empirical validation. LSTM learns long-term dependencies: it takes the concatenated function of business, marketing or financial data as input while integrating encapsulations, using wearables, smart data and social networks to capture relevant secure information with CNN in combining various search results. It is demonstrated that he learns to locate and recognize several aspects of marketing, business, and finance. The model is evaluated on the challenging tasks of the COVID-19 health crisis, electrical fraud detection in business and finance, and counterfeiting: it is shown to be more accurate than state-of-the-art of other neural networks and uses less training time.

Keywords: Big Data, CNN-LSTM-based multimodal deep learning model, Connected objects (sensors), digital technologies, digital transformation, Economic activities, Smart education, Safe information, Social networks

1. Introduction

In daily life, various business, marketing, and financial information is circulating from multimodal sources (connected objects, smart data, and multiple social media sources), some of them on all social networking channels, with generally qualitative and significant differences [1] & [2] in the information obtained. It is important to capture content [2] from connected objects, smart data, or various social media to capture a comprehensive view of an event in progress. Managers are faced with retrieving a huge amount of information shared across the web, which poses a significant challenge.

To help managers identify and filter relevant, useful shared content, several artificial intelligence systems [1] & [2] are designed to monitor the network environment. Some work is focused solely on the use of social networks as an information source, while others are simultaneously dedicated to warning [1], education [2], awareness raising [3] & [4], and digitalization [5].

Previous studies have shown that despite the investments made by several countries in integrating technology into society, the results are not promising [6, 7]. These issues have been highlighted during the recent COVID-19 pandemic as governments have been forced to move to online use at all levels of social activity (such as education, healthcare, business, etc) [5].

Designing multimodal deep learning to enhance economic activity using Big Data, high-performance computing, and education with safe information is challenging. Therefore, a new model to improve economic activities is proposed: it is based on a CNN-LSTM for learning long-term dependencies. This new approach enables the integration of artificial intelligence technologies, deep learning, and social media into multimodal deep learning for managing economic activities [3] using secure information [8, 9].

Building on previous work [2, 3, 8, 9] that forms the background for this approach, this new model combines representation training with warning and smart education while incorporating encapsulations from connected objects, smart data, and multiple social media sources.

This research study consists of monitoring the networking environment to identify relevant content of a possible upcoming economic crisis event. This could be a health crisis event such as a COVID-19 pandemic. Once the content has been retrieved and cleansed of non-informative information, it will be used to update content (warning or smart education) for citizens and managers who need to make quick and effective decisions to modify trends that would avoid major uncontrollable crises in the community (shortage of flour, milk, especially for children, drinking water, etc.) that would trigger protest marches,

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looting, sabotage, etc.

2. Background & Related Works

Online use at all levels of social activity (such as education, health, business, etc.) during the recent COVID-19 pandemic [5].

The investments made by several countries to integrate digital technology within society, the results are not promising [6, 7].

There is a lot of research on forecasting of financial time series [10]. Research on fraudfinding is common [10]. In contrast, there are only two studies on pricing [11] and only one on product placement in supermarkets. Twitter (now called X) is the first social network used in news tweeting applications [12, 13, 14]. It is followed by the NN-based crisis management model to improve alerting [1], which uses Twitter (now called X) & Facebook.

Electricity fraud detection approach [15] using a combination of Long Short-Term Memory (LSTM) methods and robust Exponential and Holt-Winters Smoothing (EHWS) methods, using sensors to improve the accuracy and efficiency of fraud detection mechanisms, while studying the fraudulent behavior of electricity consumers. It addresses the huge problem of non-technical losses (NTL), also known as electricity fraud, for electricity distribution utilities.

A total of 385,714 fake goods [16] will be confiscated by the Algerian tax authorities in 2020, according to the anti-fraud department of the Algerian tax administration [1].

There are some other works, namely LSTM [10], RNN [3], and Deep CNN-LSTM model [8, 9] to improve warning, awareness, and education in crisis events using the whole web. There are other models to improve preventing fake news, namely improving commerce and business, financial time series forecasting, fraud detection, and marketing using deep learning, big data, and high-performance computing while avoiding fake news [8, 9], followed by deep learning-based crisis management with intelligent education in environmental sustainability while avoiding false news [4], followed by context-based false news detectio, and fake news detection using CNN-BLSTM based hybrid model through context social [4]: All these models use the whole Web.

There is the first model that uses smart data & the whole web: it is the RNN-based ALE [3] that improves awareness. Other applications based on sensor technology [17, 18] are being used as the IoT advances. Thus, the study could integrate multimodal deep learning [19]. The proposed model includes multimodal deep learning design and implementation to enhance economic activities using big data, digital technologies, and education.

If the content is processed quickly and efficiently, online content contains important information that can contribute to rapid decision-making in support of affected communities. Several types of processing techniques, ranging from comparable document-based data, statistical analysis, and natural language processing to machine learning [1, 2] and computational linguistics, are developed for different purposes without fully utilizing these data, despite the existence of certain resources such as tagged data and standardized lexical resources.

Table 1: Comparing techniques and methods in models using various Sources of Capturing Content

Ref	Method	Sources of Capturing Content
[13]	Game-based learning in flood crises	Twitter
[14]	An educational objective of higher education, specifically related to	Twitter
[20]	Socio-temporal context summary	Twitter
[21]	Semi-automated AI-based classifier for crisis response	Twitter
[12]	Abstract-extractive approach to summarising contextual tweets in a crisis situation	Twitter
[1]	NN-based warning model	Facebook & Twitter
[2]	FFNN-based ALE to enhance warning and education	All the Web
[10]	LSTM-based ALE to enhance warning and education	All the Web
[3, 4, 8, 9]	CNN-LSTM-based Managing Crisis Model to enhance Warning, Awareness & Education via the Web	All the Web
[22, 23]	RNN-based model	Smart Data & All the Web
[17, 18]	Sensor-based model	Sensor
[15]	Detection of economic electricity meter anomalies and customer fraud cases	Sensor
Our New Approach	CNN-LSTM-based Multimodal DL using Sensors, Smart Data, Social Media, and Digital technologies in Economic Monitoring to improve economic activities and Fraud Detection	Connected objects (Sensors), smart data, Social & Digitalization

The shelves on which products are displayed are one of the most important resources in the retail environment, given the wide variety of products and shopping behaviors. Retailers can increase profits and reduce costs through proper management of shelf allocation and product display. Through the use of learning models in the organization of shelves in supermarkets by grouping products that are usually bought together, the research study can extract the following relationship: customers who buy product X at the end of the week, during the summer, usually also buy product Y (see Table 2). To improve the accuracy and efficiency of fraud detection mechanisms and to study the electricity consumers fraud behavior, [15] combines LSTM (Long Short-Term Memory) and EHWS techniques. It addresses the huge problem of non-technical losses (NTL), also known as electricity frauds, for electricity distribution companies. A total of 385,714 counterfeit goods will be seized by the Algerian tax authorities in 2020, according to the anti-fraud service of the General Directorate of the Algerian tax authorities [3].

[1] https://www.aps.dz/en/economy/39561-customs-over-385-000-counterfeit-goods-seized-in-2020

[3] https://www.aps.dz/en/economy/39561-customs-over-385-000-

Table 2: Economic activity examples

Economic activities	Models
Novelty fraud detection	**[24]**
Trading performance	**[25]**
Forecasting the exchange rate	**[11]**
Improving warning and education with RNN-based Model	**[22, 23]**
Improving warning and education with LSTM-based ALE	**[10]**
Improving warning and education with CNN-LSTM-based Model	**[8, 9]**
Financial time series forecasting	**[26]**
Company shares	**[27]**
Detection of economic electricity meter anomalies and customer fraud cases	**[15]**
385,714 counterfeit products were confiscated by the Algerian tax authorities in 2020, according to the anti-fraud service of the Algerian tax administration[2]	**[16]**

Table 3: Integrating digital technology into Economic activity

Digitalizations	Models
Online use at all levels of social activity (such as education, health, business, etc.) during the recent COVID-19 pandemic	**[5]**
Investments made by several countries in integrating technology into society Results are not promising	**[6, 7]**
CNN-LSTM-based Multimodal DL using Sensors, Smart Data, Social Media, and Digital technologies in Economic Monitoring to improve economic activities and Fraud Detection	**New Approach**

In the unsupervised learning category, class labels are either unknown or assumed to be unknown, and clustering techniques are used to find (i) distinct clusters containing fraudulent samples, or (ii) distant fraudulent samples not belonging to any cluster where all clusters contain genuine samples, in which case it is treated as an outlier detection problem.

The supervised learning approach takes known class labels and constructs a binary classifier. Fraud (including cyber fraud) detection is increasingly threatening and fraudsters always seem to be several steps ahead of organizations in finding new loopholes in the system and bypassing them effortlessly.

Smart Data always combines analysis of this data with human intelligence to provide comprehensive, synthesized, and useful business information (see Table 5).

This education consists of constantly repeating certain advice on all information channels, websites, and social and networking media to maximize situational awareness.

3. New Model for Monitoring Economic Environment

The new multimodal CNN-LSTM model is presented. The LSTM layer handles temporal correlations. Its enhancement with convolutional structures in both input and output states will solve the problem. By stacking several CNN-LSTM layers

counterfeit-goods-seized-in-2020

Table 4: Domains of application for smart data

Applications Domains	Limitations
Financial services	Fraud detection and prevention
Retail	Allowing brands to: - Analyzing customer sentiment - Offering personalized and contextual promotions
Telecommunications industry	Possibility of - Better allocating bandwidth based on real-time needs, and - diagnosing antenna condition
Understand the market conditions	Costs
Increased productivity	Security and Privacy Concerns
Fraud detection	/
Production	Preventive maintenance
Healthcare	- Monitoring patient vital signs, - Reducing readmission rates
Oil industry	- Proactive repairing infrastructures, and - Balancing the power delivered according to consumption
Public sector	- Detecting and preventing intrusion attempts on the network - & Predicting epidemic risk
Transport sector	Possibility of detecting risky conduits

and building a coding prediction structure, the research created a network model for these spatiotemporal prediction problems. The goal of crisis prediction is to use the previously observed sequence of multimodal sources to prevent an event in a local area, such as Algiers, London, or Paris. From an automatic learning perspective, this is a spatiotemporal prediction problem. Assume a dynamic system represented by a MxN grid. Each cell of the grid contains P time-varying measures (word, bias). Each observation can be represented by a tensor X of RPxMxN, where R denotes the domain of observed features. If periodic observations are recorded, the study has a sequence of tensors $X_1, X_2, ..., X_t$. This is formulated by Eq. (1) as the likeliest series of length K from all previous series of length J, including the current series:

$$\hat{Y}_{t+1}, \cdots, \hat{Y}_{t+K} =$$

$$\mathbf{arg} \max_{\mathbf{X}_{t+1},...,\mathbf{X}_{t+K}} \left\{ p(\mathbf{X}_{t+1}, ..., \mathbf{X}_{t+K} \mid \mathbf{Y}_{t-J+1}, ..., \mathbf{Y}_{t-J+K}) \right\} \quad (1)$$

Consider a hidden-layer CNN-LSTM-based multimodal deep learning model with input content formulated by the equation:

$$e = (w_1, ..., w1, ..., w_n) \quad (2)$$

containing words W, each from a finite vocabulary **V**, i.e. the content set is issued from connected objects, smart data, and multiple social networks, and giving as output the relevant text $\mathbf{e_k}$.

Full connectivity makes them susceptible to overfitting, as in (3).

$$\forall n \in [1, 2n_c^{[l]}]Conv(a^{[l-1]}, K_{x,y}^{(n)}) =$$

$$\phi^{[l]}\Big(\sum_{i=1}^{n_H^{[l-1]}} \sum_{j=1}^{n_W^{[l-1]}} \sum_{k=1}^{n_C^{[l-1]}} K^{(a)}_{i,j,k} * a^{[l-1]}_{x+i-1,y+j-1,k} * b_n^{[l]}\Big) \tag{3}$$

$$Dim(Conv(a^{[l-1]}, K^{(n)}) = (n_H^{[l]}, n_W^{[l]})$$

The formula for the current state is as follows

$$h_s = f(h_{s-1}, x_s) \tag{4}$$

Applying activation function tanh:

$$h_s = \tanh(\sigma_{hh} * h_{s-1} + \sigma_{xh} * x_s) \tag{5}$$

where

- $*$ denoting multiplication by elements.
- ω the function of the logistic sigmoid.
- i, f, and j are, respectively, input, forget, and output gates
- c the vector of cellular activation, the same size as hidden vector h, at level k.

The formula of current state follows as:

$$h_s = f(h_{s-1}, x_s) \tag{6}$$

Applying the function of activation tanh:

$$h_s = \tanh(\sigma_{hh} * h_{s-1} + \sigma_{x_h} * x_s) \tag{7}$$

where

- σ is weight.
- h denotes a sole vector that is hidden,
- σ_{hh} weight of the last hidden state,
- σ_{xh} weight of current input state,
- tanh the function of activation introducing an activation of non-linearity squashing to [-1,1].

Denoting output:

$$y_s = \sigma_{hy} * h_s \tag{8}$$

where

- y_s output state, and
- σ_{hy} weight of output state.

Figure 3 shows the overview of LSTM, gates, activation functions.

Let the following formula:

$$\forall i \in [1, N]\ e_i \in \mathbf{C^n} = \mathbf{E}\ \&\ \mathbf{e_i} = (w_{i1}, ..., w_{in})$$

containing words from the word set **W**. Each word built from a finite vocabulary **V**, the inclusion of the content of the source message is relevant to at least one tag or hash:

$$\exists j \in [1, M]\ such\ as\ h_j \in \mathbf{H}$$

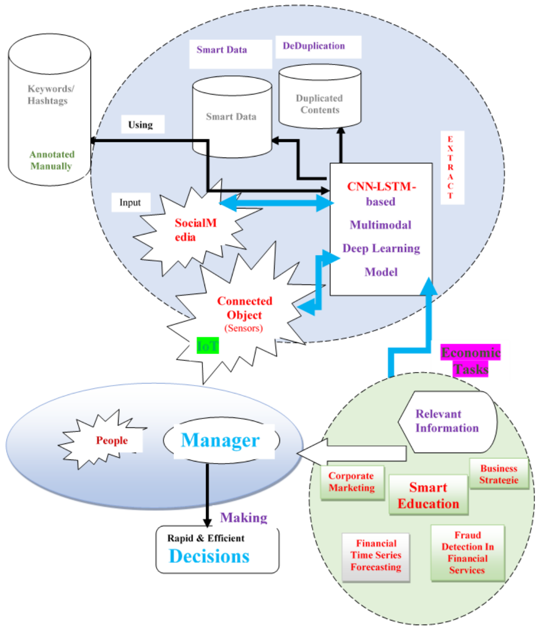

Figure 1: Functioning of the multimodal hybrid model based on Deep CNN-LSTM.

It wants to learn a generic space as formulated by this formula:

$$\mathbf{E_K} = \max_{k\in[1,K]} \left\{ e_k \right\}$$

normalize the differences in equation (4):

$$\mathbf{E_K} = [\mathbf{E} - \mathbf{RDF}]\ where\ \mathbf{RDF} = [\mathbf{R} + \mathbf{D} + \mathbf{F}] \tag{9}$$

where

- **R**, **D** & **F** denote respectively the set of duplicate re-tweets, duplicate contents and false alerts.
- Figure 4 shows how the multimodal hybrid model based on Deep CNN-LSTM works.
- The CNN-LSTM-based multimodal deep learning model can explain the transformation of e_i into e_k by

$$\max_{k\in[1,K]} \left\{ e_i \to e_k\ =\ e_i\ such\ e_i\ is\ relevant\ for\ h_j\ \&\ w_l \right\}$$

$$with\ \ i\ \in [1, N]\ and\ e_i \notin [\mathbf{R} + \mathbf{D} + \mathbf{F}]$$

Where

- **R**, **D**&**F** denote the duplicate retweets, duplicate content, and false alert set respectively.

The goal is then to maximize the size of the set **K** of $\mathbf{E_K}$. Figure 3 shows the architecture of the multimodal Deep CNN-LSTM-based economic monitoring model.

A student can use this tool in three modes. Novice mode allows him to use a complete set of automated design and learning

Table 5: Comparing techniques and methods in models with education

Models	Educational Methods
[12]	An abstractive-extractive approach to situational tweet summarisation in crisis events
[3, 4, 8, 9]	Managing crises through education, based on CNN-LSTM
[14]	The importance of education during crises
[13]	Opportunities and challenges of education programs
[2]	FFNN-based model with smart education
[22, 23]	RNN-based ALE to enhance awareness
[10]	LSTM-based model with smart education
Our New Approach	CNN-LSTM-based Multimodal DL using Sensors, Smart Data, Social Media, and Digital technologies in Economic Monitoring to improve economic activities and Fraud Detection

tools, like [2]observing different programs at work, experimenting, and gradually learning from his experiences, observations, and mistakes. Using the novice mode, he can ask it at any time to generate the next step. Analyzing knowledge provides the optimal stage and list of relevant operations. Unsatisfied with the suggested operation, the trainee can choose any operation from adaptive menus. In the online tutorial mode, at any time during the training, the trainee has a menu to access all previous courses, such as:

- Present any previously learned concept,
- Demonstrate all examples and analyze problems explained or solved. This mode gives access to the course material as a reference [2, 3, 10]: thereby
- supporting online help based on examples. Training in commercial, financial, or fraud prediction [3] consists of:
- Advice to always pay attention to the quality of the products to be consumed, with the reinforcement of preventive hygiene, namely
  - Check the quality, price, manufacture, and expiry date of each product
  - Strengthening preventive hygiene,
  - check that each product has indications visible on its packaging, how store it, manufacturing date, expiry date, batch number
- respect the conditions for using the product,
- Do not use anything without reading the instructions,
- End anarchic and uncontrolled buying,
- Promoting health and hygiene and
- Avoid inappropriate use (social, cultural).

This multimodal deep-learning model aims to standardize good service practices, developing essential skills, knowledge and expertise for companies and business people in general during a crisis, including:

- Skills appropriate to the sales service position or function on demand,
- Unique skills with a focus on their level, not the role,
- Specific roles and skills,
- Graded skills required according to managing the local economic process,
- Skills based on different target groups, and
- Application of transversal skills to business staff.

Information security relies on confidentiality, integrity, availability, non-repudiation (multiparty transactions) and authentication. Unfortunately, social media lacks information security. Information security is social media's main business and financial concept. There are two successive ways of doing this. First, the security is against content search from a multimodal deep learning approach with connected objects, smart data, and multiple social media sources [1, 2, 3, 10]. Then, security from false information [3] (see Table10).

Table 6: Comparison of methods and techniques in fake news prevention models.

Models	Methods
[9]	Model economic environment monitoring to enhance economic activities in sustainability using deep learning with economic education from various sources
[8]	Modelling CNN-LSTM-based model for economic monitoring and crisis forecasting in sustainability
[3]	Enhancing trade, business, financial time series prediction, fraud detection and marketing with DL, big data, HPC with safe information
[4]	Smart education with DL and social networks to manage sustainability with safe news
Our New Approach	CNN-LSTM-based Multimodal DL using Sensors, Smart Data, Social Media, and Digital technologies in Economic Monitoring to improve economic activities and Fraud Detection

4. Results and Discussion

Table 11 shows the comparison between methods and techniques in models to Sustainability.

Table 7: Comparison of sustainable methods and techniques.

Models	Methods
[9]	Using deep learning with economic education from various sources to model the monitoring of the economic environment to strengthen economic activities in sustainability
[8]	Modeling CNN-LSTM-based model for economic monitoring and crisis forecasting with safe news in sustainability
Our New Approach	CNN-LSTM-based Multimodal DL using Sensors, Smart Data, Social Media, and Digital technologies in Economic Monitoring to improve economic activities and Fraud Detection

Table 8: COVID-19 damage estimation: (affected, dead and cured) since May 31^{st}, 2020, in Algeria.

Period	Affected	Death	Healed
05/31/2020	9,394	653	/
06/31/2020	10,050	698	/
07/21/2020	10,050	1,100	/
08/24/2020	10,050	1,446	/
10/01/2020	10,050	1,783	36,958
11/10/2020	10,050	2,077	42,626
11/12/2020	10,050	2,093	42,980
03/16/2021	115,440	3,040	39,994

Table 9: From March 2020 to date, examples of relevant content of COVID-19 in Algeria

Models	COVID-19 in Algeria
Support Vector Machine (SVM) [17, 30]	2364
Neural Network (NN) via Twitter & Facebook [1]	2792
Feed-forward Neural Network (FFNN) via All the Web [2]	2971
Recurrent Neural Network (RNN) via Smart Data & All the Web [3]	2495
LSTM via Smart Data & All the Web [3]	2804
Hybrid of Deep CNN-LSTM via All the Web [17, 18]	2634
Our New Approach via Sensor, Smart Data & All the Web	4998

Table 10: Examples of relevant content of electricity fraud detection in Algeria

Models	Electricity fraud detection in Algeria
Support Vector Machine (SVM) [17, 30]	2444
Neural Network (NN) via Twitter & Facebook [1]	2812
Feed-forward Neural Network (FFNN) via All the Web [2]	3161
Recurrent Neural Network (RNN) via Smart Data & All the Web [3]	2515
LSTM via Smart Data & All the Web [3]	2924
Hybrid of Deep CNN-LSTM via All the Web [17, 18]	2664
Our New Approach via Sensor, Smart Data & All the Web	6118

A total of 385,714 fake goods will be confiscated by the Algerian tax authorities in 2020, according to the anti-fraud department of the Algerian tax administration [4].

Table 12 is an example COVID-19 assessment. Examples of relevant content for a set of hashtags and keywords from multiple sources for the Algerian forest fire of August 16^{th} – 19^{th}, 2021. Each message is represented by a sequence of transactions T= $(t_1, ..., t_n)$. COVID-19 for Algeria, from March 2020 until today, for a set of hashtags and keywords for multimodal sources.

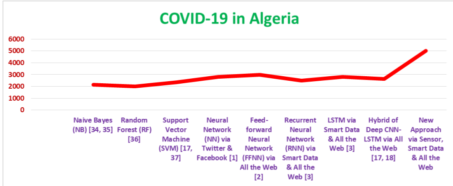

Figure 2: Examples of Relevant Content of COVID-19 in Algeria from March 2020 to date.

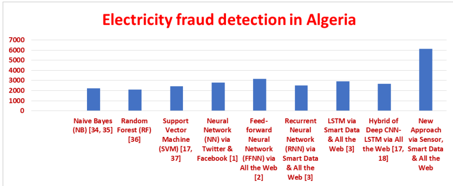

Figure 3: Examples of Relevant Content of electricity fraud detection in Algeria from March 2020 to date.

Table 11: From March 2020 to date, examples of relevant content of counterfeiting in Algeria

Models	Counter-feiting in Algeria
Support Vector Machine (SVM) [17, 30]	2464
Neural Network (NN) via Twitter & Facebook [1]	2972
Feed-forward Neural Network (FFNN) via All the Web [2]	3171
Recurrent Neural Network (RNN) via Smart Data & All the Web [3]	2513
LSTM via Smart Data & All the Web [3]	3004
Hybrid of Deep CNN-LSTM via All the Web [17, 18]	2804
Our New Approach via Sensor, Smart Data & All the Web	6198

Table 13 and Figure 5 show the comparison between the results obtained in this approach with the previous results of the study, namely Neural [1], Neural [2], RNN [3], LSTM [10] and CNN-LSTM [8, 9] like this work, for COVID-19 in Algeria.

5. Conclusion and Perspectives

In conclusion, using a CNN-LSTM-based multimodal deep learning model for economic monitoring and crisis forecasting can fundamentally change how people, especially managers, improve economic activities. The study investigated integrating IoT, big data, HPC, and multiple social media technologies for economic monitoring and crisis forecasting. The results provide insight into the perceived impact on warning, effectiveness, user satisfaction, and favorable attitudes. It shows that multimodal deep learning based on CNN-LSTM can improve economic activities and crisis prediction. Citizens and administrators may conduct economic transactions in full confidence, in a regulated and secure environment where information will be protected. The results of the research agree with the idea

[4] https://www.aps.dz/en/economy/39561-customs-over-385-000-counterfeit-goods-seized-in-2020

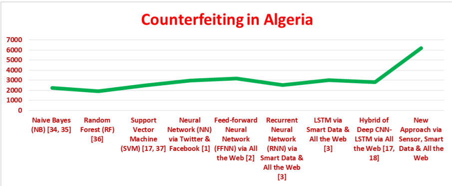

Figure 4: Examples of Relevant Content of Counterfeiting in Algeria from March 2020 to date.

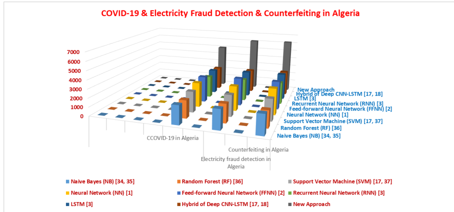

Figure 5: Comparing Examples of Relevant Content of Health crisis of COVID-19, electricity fraud detection in business and finance, and counterfeiting in Algeria.

Table 12: Comparing Examples of Relevant Content of Health Crisis of COVID-19, Electricity Fraud Detection in Business and Finance, and Counterfeiting in Algeria

Models	COVID-19 in Algeria	Electricity fraud detection in Algeria	Counter-feiting in Algeria
Support Vector Machine (SVM) [17, 30]	2364	2444	2464
Neural Network (NN) via Twitter & Facebook [1]	2792	2812	2972
Feed-forward Neural Network (FFNN) via All the Web [2]	2971	3161	3171
Recurrent Neural Network (RNN) via Smart Data & All the Web [3]	2495	2515	2513
LSTM via Smart Data & All the Web [3]	2804	2924	3004
Hybrid of Deep CNN-LSTM via All the Web [17, 18]	2634	2664	2804
New Approach via Sensor, Smart Data & All the Web	4998	6118	6198

Table 13: Overview of Comparison between the results obtained in this approach

Models	RMSE	MAE	R^2
Support Vector Machine (SVM) [17, 30]	77,000	60,000	0,190,357,14
Neural Network (NN) via Twitter & Facebook [1]	91,000	113,333	0.317,500
Feed-forward Neural Network [1]	105,500	130,000	0.358,208,66
Recurrent Neural Network (RNN) via Smart Data & All the Web [3]	14,000	12,666	0.456,043,96
LSTM via Smart Data & All the Web [3]	112,000	106,666	0.491,578,93
Hybrid of Deep CNN-LSTM via All the Web [17, 18]	103,000	103,333	0.499,959,51
Our New Approach	103,400	773,333	0,952,846,56

that this new multimodal deep learning approach for economic monitoring and crisis forecasting systems can bridge the gap between theoretical understanding and practical implementation.

Although the research highlights the possible benefits of this new approach, the validation of this new system will require feedback on the use of this new approach for economic surveillance and crisis prediction using descriptive statistics through sampling. This will be the role of future work. The study presented an ad hoc real-time multimodal deep learning model. It is based on a CNN-LSTM for educational warning to prevent events. These include data from networked objects, smart data, and multiple social media. Such an approach is particularly useful for improving economic activity. The community can be alerted to potential events, informed of status, and request help. With data, the content is written informally, with no syntax, logic, or noise. There are also indicators of pollution, temperature, humidity, etc. Therefore, it may be biased towards certain domains. Multimodal deep learning model features are developed by analyzing specific event content.

In future works, this research study has several potential applications. Validating relevant information (avoiding misuse), using different languages (including Arabic and French), and extracting the most useful to allow the community to live easily and comfortably, saving the lives of those stuck under the rubble, avoiding any complications, difficulties, material, social or even professional. The real-time paradigm could be extended by using big data to search for information about past events, validating a possible warning to keep people safe. In addition, quantitative research involves surveying a sample of clients and managers to gauge their views on how to improve economic monitoring and crisis forecasting through education.

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Declarations

*Funding does not require any funding: this work is within the framework of research

* Conflicts of interest: All metrics are the property of the LMA laboratories of the University of Bejaia, Algeria

*Availability of data and material: The data supporting this study's findings are available from the corresponding author. However, restrictions apply to the availability of these data, which were used in the LMA laboratory from the University of Bejaia, in Algeria for the current study, and so are not publicly available. Data are however available from the corresponding

author upon reasonable request and with permission of Managers of the LMA laboratory.

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